Python Assignment 3: While Loop, For loop and Function

Step 1: Number Guessing Game

```python
import random
secret_number = random.randint(1, 10)
attempts = 3
while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))
    if guess < 1 or guess > 10:
        print("Your guess is out of range. Please guess a number between 1 and 10.")
        continue
    if guess == secret_number:
        print("Congratulations! You guessed the correct number.")
        break
    elif guess > secret_number:
        print("Too high. Try again.")
    else:
        print("Too low. Try again.")
    attempts -= 1
else:
    print("Better luck next time!")
    print("The correct number was:", secret_number)
```

```
Guess the number (between 1 and 10): 2
Too low. Try again.
Guess the number (between 1 and 10): 8
Too high. Try again.
Guess the number (between 1 and 10): 1
Too low. Try again.
Better luck next time!
The correct number was: 3
```

Step 2: For Loop – Multiplication Table Generator

```python
number = int(input("Enter the number for which you want the multiplication table: "))
for i in range(1, 11):

    result = number * i

    print(number, "x", i, "=", result)
```

```
Enter the number for which you want the multiplication table: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

Step 3: Function – BMI Calculator

```
def calculate_bmi(weight, height):

    bmi = weight / (height ** 2)

    return bmi

weight = float(input("Enter your weight in kg: "))

height = float(input("Enter your height in meters: "))

result = calculate_bmi(weight, height)

print("Your BMI is:", round(result, 2))
```

```
Enter your weight in kg: 45
Enter your height in meters: 85
Your BMI is: 0.01
```